Lab 12 - Infill Patterns - Zigzag vs Grid Strength Test

Course: Innovating Product Development with 3D Printing and Additive Manufacturing (TGS-2022017597) Topic 03: Process Improvement for AM TSC objective: A5 - evaluate infill pattern and density as a quality-improvement technique

Goal

Print cube10.stl with zigzag 20% and grid 30% infill, tune the top layers, then compare the two cubes' stiffness by hand.

What you'll build

Two printed cubes with different infill patterns/densities and a hands-on strength comparison.

Tools: Ultimaker Cura, cube10.stl, Optimize1-4 model set, FDM printer

Steps

1. Open cube10.stl and set infill pattern to Zigzag at 20% density.

```
Infill > Pattern - Zigzag, Density - 20%
```

2. Set top layers to 4 so the roof closes cleanly over the infill.

```
Top/Bottom > Top Layers - 4
```

3. Slice, save and print the first cube.

```
Slice > Save to Disk > print
```

4. Change the pattern to Grid at 30% density and re-slice; note the time difference.

```
Infill > Pattern - Grid, Density - 30%
```

5. Print the second cube.

```
Save to Disk > print
```

6. Compare the cubes: squeeze the walls, press the top faces, and note pillowing or gaps over the infill.

```
Hand test: wall flex + roof sag
```

7. Load optimize-1.stl through optimize-4.stl, orient each for minimum support, and record Cura's estimated time/material before choosing the most efficient printable variant.

```
Open four models > Auto Orient/Rotate > Slice > record estimates
```

Evidence to capture

- Optimisation scorecard for all four supplied variants: orientation, support volume, time and material.

- A justified selection of the best variant/settings for the stated business need.

Test it

The grid 30% cube is stiffer and better supported, while your four-variant scorecard shows that geometry and orientation can outweigh infill choice; your final recommendation is backed by time, material and support evidence.

Files in this folder

- cube10.stl - Core sample model used by the lab procedure.
- optimize-1.stl - Complex baseline part for printability and support-cost review.
- optimize-2.stl - Wide low-profile variant for build-orientation comparison.
- optimize-3.stl - Twisted variant for support and toolpath comparison.
- optimize-4.stl - Compact variant for rapid slicing and material-use comparison.